KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2021

EE 2113

(Electrical Machines)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

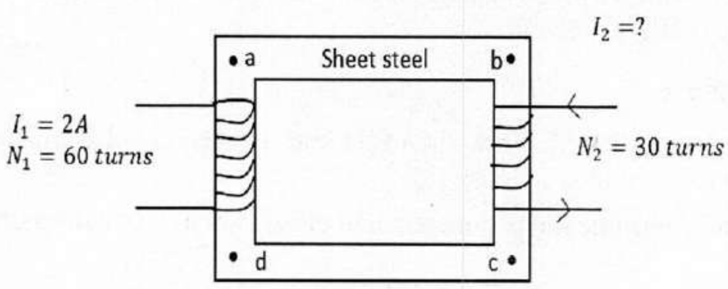
SECTION – A

1(a). Define the following terms : 09
i) Relative permeability, ii) Faraday's law of electromagnetic induction, iii) Magnetization force

1(b). State and Explain Ampere's Circuital laws of magnetic circuit with necessary illustrations. 10

1(c). Write down the differences between electric and magnetic circuit. 06

1(d). Determine the secondary current I_2 for the transformer shown in figure, if resultant clockwise flux in the core is $1.5 \times 10^{-5} Wb$. 10



Area (throughout) = $0.25 \times 10^{-3} m^2$

$l_{abcda} = 0.15m$

For sheet steel B-H data are given in the table below

B_2 (Wb/m^2)	H(At)
0.05	10
0.1	20
0.2	40

2(a). Describe the voltage build up process of a DC shunt generator. Mention the factors affecting the voltage bulid up process. 10

2(b). Defin ecritical resistance. For DC shunt series, short shunt and long shunt compund generators, write down the (i) emf equation, (ii) Armature current equation. 10

2(c). A four pole, long shunt DC compound generator supplies 250 V at 50A in the load. The generator has Armature, Series and shunt field resistances of 0.5Ω, 0.3Ω and 200Ω, respectively. Find the following: 15
i) The generated emf and armature current.
ii) If total number of armature conductor is 500, and rotated in 1500 rpm what will be the flux required to generate the voltage as in (i).

3(a). What is back emf? Explain the significence of back emf in motor. 10

3(b). " Back emf self-regulates the motor" – Justify the statement. 10

3(c). A belt-driven $100kW$,shunt generator running at $500rpm$ on $220\ V$ bus bars continues to run as a motor when belt breaks, then taking $10kW$. What will be its speed? Given: armature resistance $= 0.025\Omega$, field resistance $= 60\Omega$ and contact drop under each brush $= 1V$. Ignore armature reaction. 15

4(a).	" The speed can be controlled above and below the no-load speed only by flux control method" -explain the statement.	10
4(b).	Why starter is required for DC motor? Explain the working of four point starter.	10
4(c).	List the difference between DC generator and DC motor.	07
4(d).	For converyer belt application in industry, which DC motor do you use? Draw the characteristic curves of that motor.	08

SECTION – B

5(a).	"The net flux passing through the core is approximately the same as no load". Explain it.	10
5(b).	Why transformer is necessary in power transmission. Draw and explain the parameters of single phase transformer.	08
5(c).	The parameters of a $2300/230\ V$, $50Hz$ transformer are given below : $R_1 = 0.286\Omega,\ R_2 = 31.9\Omega,\ R_0 = 250\Omega,\ X_1 = 0.73\Omega,\ X_2 = 73\Omega,\ X_0 = 1250\Omega$. Solve the exact equivalent circuit with normal voltage across the primary. Consider the secondary load impedance is $0.387 + i\ 0.29$.	17
6(a).	Why transformer rating is given in kVA? Draw the exact and approximated equivalent circuit of a transformer with referred to secondary side.	10
6(b).	Explain briefly the process to determine the parameters and efficiency of a transformer.	15
6(c).	Define the following terms i)Transformer ration, ii) Core Ion, iii) Copper Ion	10
7(a).	Why induction motor is called rotating transformer?	05
7(b).	Show that the maximum value of the rotating flux is constant in induction motor.	12
7(c).	Draw and explain the torque -slip (speed) characteristics of a slip ring induction motor.	10
7(d).	A 3-Phase induction motor is wound for 4 pole and is supplied from 50Hz system. Calculate (i) Synchronous speed, (ii) Rotor speed when the slip 4% and (iii) rotor frequency when rotor runs at$1200\ rpm$.	08
8(a).	Draw and explain the vector diagram of an alternator having lagging load.	08
8(b).	Write down the advantages and disadvantages of alternator.	06
8(c).	In a 50 kV star connected 440 V, 3-phase 50Hz alternator, the effective armature resistance per phase is 0.25Ω. The synchronous reactance and leakage reactance per phase are 3.2 Ω and 05 Ω respectively. Determine at rated load and unity power factor condition i)Internal e.m.f E_a, ii) No load e.m.f E_0	10
8(d).	Define voltage regulation. Draw and explain the voltage regulation graph for i) leading, ii) lagging, and iii) unity power factor load.	11

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2021
Math 2113
(Linear Algebra and Vector Analysis)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Find the condition on a, b and c so that the system of equations 12

$$x + 2y - 3z = a$$
$$2x + 6y - 11z = b$$
$$x - 2y + 7z = c$$

in unknowns x, y and z is consistent. Then what type of solutions exists for this system?

1(b). Identify which of the following system has non-trivial solution and solve those, if any. 17

(i) $x + 3y - 2z = 0$, $x - 8y + 8z = 0$, $3x - 2y + 4z = 0$

(ii) $x + 2y - 3z = 0$, $2x + 5y + 2z = 0$, $3x - y - 4z = 0$

(ii) $x + 2y - z = 0$, $2x + 5y + 2z = 0$, $x + 4y + 7z = 0$, $x + 3y + 3z = 0$

1(c). Find all matrices $\begin{pmatrix} x & y \\ z & w \end{pmatrix}$ which commute with $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. 06

2(a). If A= $\begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$, compute A^{-1} by elementary row transformations and hence find B using A^{-1} so that AB = $\begin{bmatrix} -1 & 3 & 2 \\ 0 & 1 & 3 \\ 1 & 3 & -1 \end{bmatrix}$. 13

2(b). Determine the rank of the matrix A= $\begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 3 \\ 3 & 6 & 2 & -6 & 5 \end{bmatrix}$ by reducing it to row cauonical form and obtain its normal form. 13

2(c). Prove that $u = (1, 1, 2)$, $v = (1,0,1)$ and $w = (2, 1, 3)$ do not span the vector space $\mathbb{R}^3$. 09

3(a). If A and B are orthogonal matrices, show that A^T, A^{-1} and AB are also orthogonal. 07

3(b). Determine whether the vectors $u = (1, -2, 3)$, $v = (5, 6, -1)$ and $w = (3, 2, 1)$ are linearly independent or dependent set. 10

3(c). Determine whether or not the following vectors form a basis for the vector space $\mathbb{R}^3$: 18

(i) (1, 1, 1), (1, 2, 3) and (2, −1, 1),
(ii) (1, 1, 2), (1, 2, 5) and (5, 4, 3).

4(a). Find the dimension of the subspace W of $\mathbb{R}^4$ generated by (1, 4, −1, 3), (2, 1, −3, −1) and (0, 2, 1, −5). 08

4(b). Find the dimension for the solution space of the system $x + y + z = 0$, $3x + 2y - 2z = 0$, $4x + 3y - z = 0$, $6x + 5y + z = 0$. 10

4(c). Let V be the space of 2 × 2 matrices over $\mathbb{R}$ and W be the subspace generated by $\begin{pmatrix} 1 & -5 \\ -4 & 2 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ -1 & 5 \end{pmatrix}$, $\begin{pmatrix} 2 & -4 \\ -5 & 7 \end{pmatrix}$ and $\begin{pmatrix} 1 & -7 \\ -5 & 1 \end{pmatrix}$. Find a basis and dimension of W. 07

4(d). Define Euclidean n-space. State Cauchy-Schwarz inequality. Find the equation of the hyperplane H in $\mathbb{R}^4$ passes through P = (3, −2, 1, −4) and is normal to (2, 5, −6, −2). 10

SECTION – B

5(a). Describe the parametric curve represented by the equations 12

(i) $x = 1 - t,\ y = 3t, z = 2t$

(ii) $x = acost, y = asint, z = ct.$

5(b). Interpret the derivative of vector-valued function with graphs. Find the parametric equation of the tangent line to the circular helix $x = cost,\ y = sint,\ z = t.$ 12

5(c). Find the normal and binormal for the circular helix $x = acost, y = asint, z = ct$ where $a > 0$. 11

6(a). Describe the graph of the following functions in an xyz coordinate system 12

(i) $f(x,y) = 1 - x - \frac{1}{2}y,$ (ii) $f(x,y) = \sqrt{1 - x^2 - y^2},$ (iii) $f(x,y) = -\sqrt{x^2 + y^2}.$

6(b). Define level curves. Sketch the contour plot $f(x,y) = 4x^2 + y^2$, using level curves of height k = 0, 1, 2, 3, 4, 5. 12

6(c). Find the partial derivative of the following functions: 11

(i) $z = e^x siny + e^y cosx$ (ii) $z = ln(x^2 + y^2) + 2tan^{-1}\left(\frac{y}{x}\right).$

7(a). Using line integration show that $\bar{F} = 3x^2y\hat{\imath} + (x^3 + 2yz)\hat{\jmath} + y^2\hat{k}$ is conservative force field. Find the amount of work done in moving an object in the field from (1, −2, 1) to (3, 1, 4). 12

7(b). Evaluate $\int_C (y - sinx)dx + cosx\ dy$ by using Greeu's theorem in place where C is the triangle OAB with vertices O (0, 0), A $(\frac{\pi}{2}, 0)$ and B $(\frac{\pi}{2}, 1)$, respectively in the anticlockwise direction. 11

7(c). Evaluate $\iint_S \bar{F}.\bar{n}\ ds$ where $\bar{F} = y\hat{\imath} + 2x\hat{\jmath} - z\bar{k}$ and S is the surface of the plane $2x + y - 6 = 0$ in the first octant cut off by the plane z =6. 12

8(a). Explain the divergence theorem geometrically. Using divergence theorem, find the outward flux of the vector field $F(x,y,z) = x^3i + y^3j + z^2k$ across the surface of the region enclosed by the circular cylinder $x^2 + y^2 = 9$ and the planes $z = 0$ and $z = 2$. 18

8(b). State the Stokes' theorem. Find the work performed by the force field $F(x,y,z) = x^2i + 4xy^3j + y^2xk$ on a particle that traverses the rectangle C in the plane $z = y$ where $0 \le x \le 1, 0 \le y \le 3$. 17

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2021
ME 2113
(Statics and Solid Mechanics)

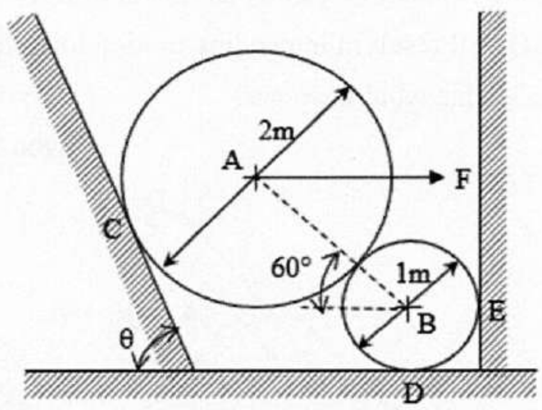
Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

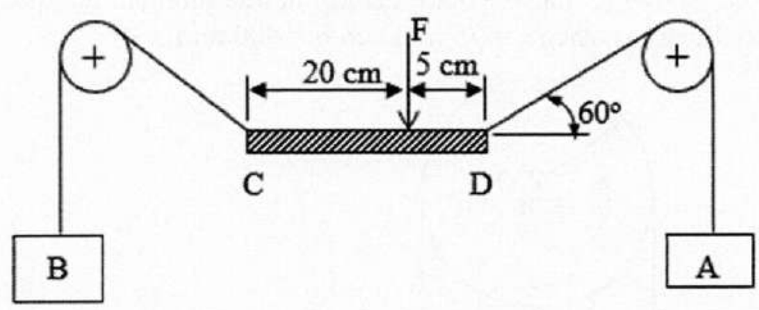
SECTION – A

1(a). State and explain the parallelogram law of forces 05

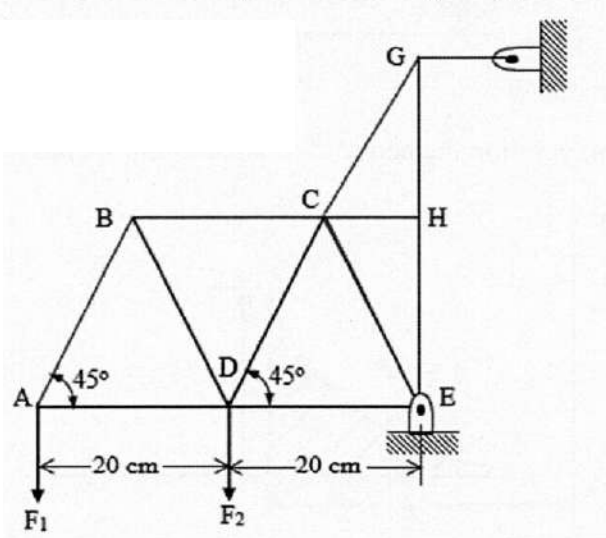
1(b). Two spheres are at rest against smooth surfaces as shown in figure. Sphere A weighs 1450 kg and sphere B weighs 180 kg. Let, F = 460 kg and $\theta = 90°$. Find the reactions at C, D and E. 15



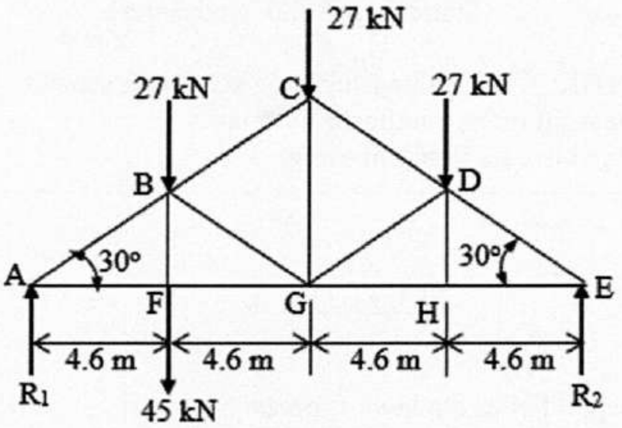
1(c). In the following figure, CD is rigid and weightless body. Let, F = 670 N, the pegs are smooth and the cable is weightless and flexible. Determine the weight of A and B if the bodies are in equilibrium and CD remains horizontal. 15



2(a). In figure shown, let $F_1 = 500$ kg and $F_2 = 2500$ kg. Find the external reactions (components of E) and the loads on the members CG, CH and EH by using the method of section. 17



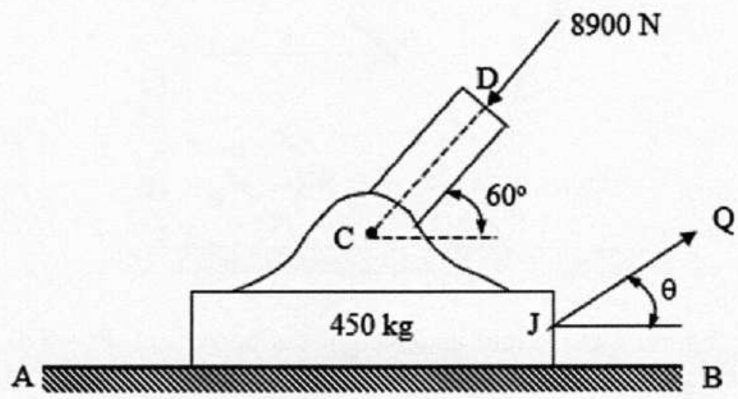
2(b). For the truss shown in figure, find the external reactions and find the loads on the members CD, DG and GH by a single section. 18



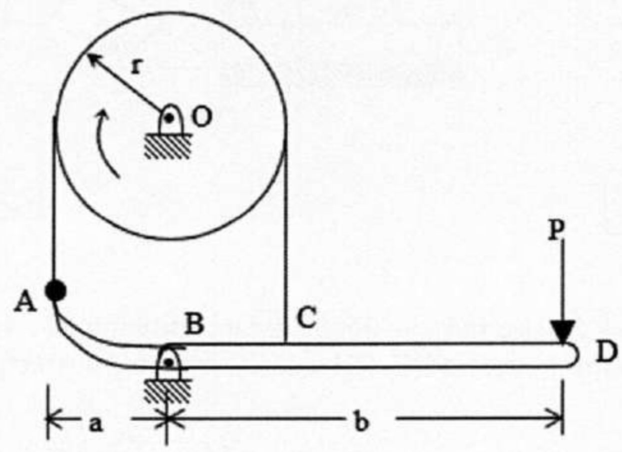
3(a). A 450 kg body is resting on a rough horizontal surface for which f = 0.25. A compressive member CD transmits 8900 N load to the body as shown in figure. 17

(i) If $\theta = 0°$, what force Q will result in impending motion toward the right?

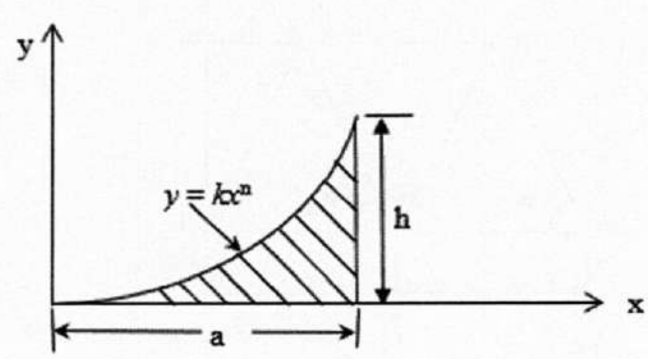
(ii) If Q was equal to zero, what would happen?



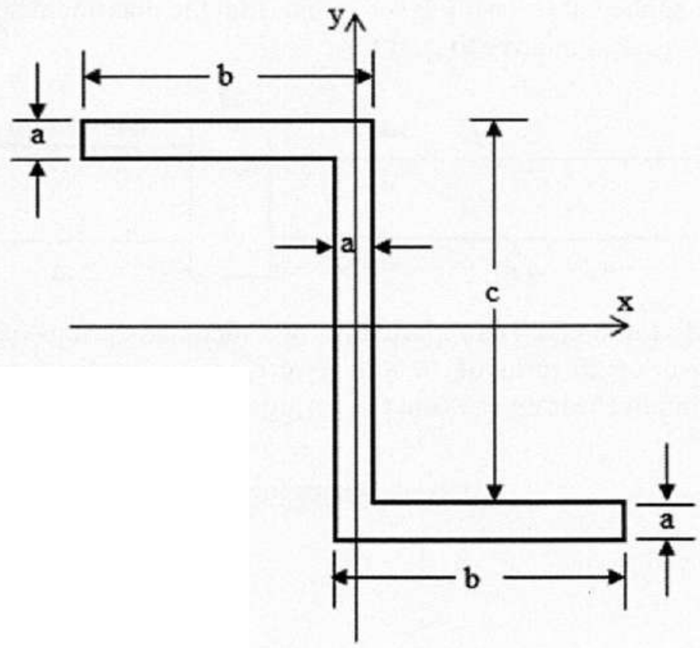
3(b). A brake drum of radius r = 150 mm is rotating clockwise when a force P of magnitude 75 N is applied at D. Knowing that f = 0.25, determine the moment about O of the friction forces applied to the drum when a = 75 mm and b = 400 mm. 18



4(a). Determine by direct integration the centroid of a parabolic spandrel as shown in figure. 17



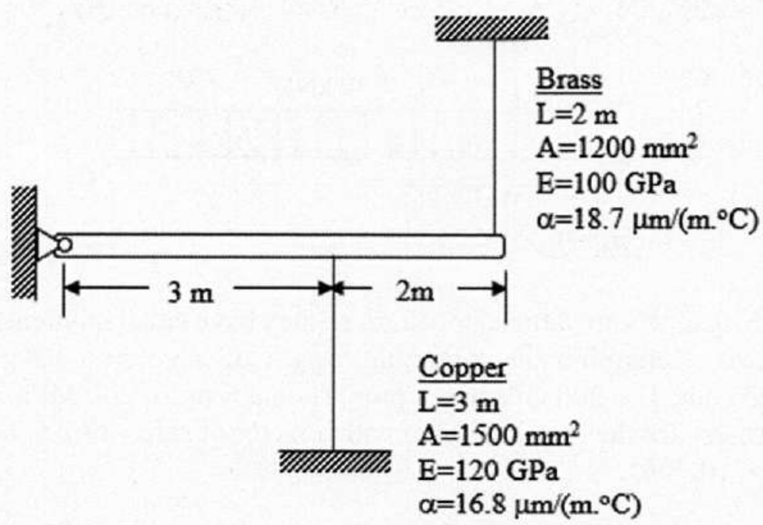
4(b). In the standard structural steel Z section shown in figure, let a = 19 mm, b = 90 mm and c = 150 mm. Find the $\bar{I}_x$ and $\bar{I}_y$. 18



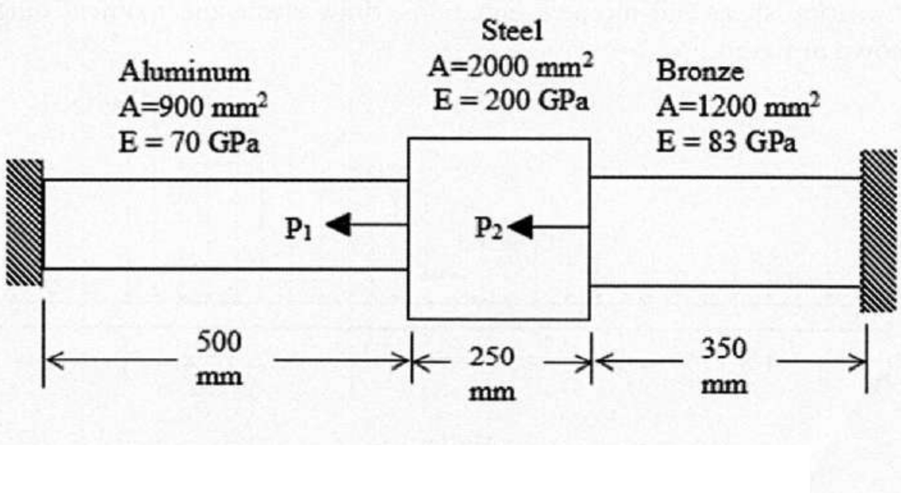
SECTION – B

5(a). State Hook's law. Draw stress Vs strain diagram for mild steel and identify different points. 05

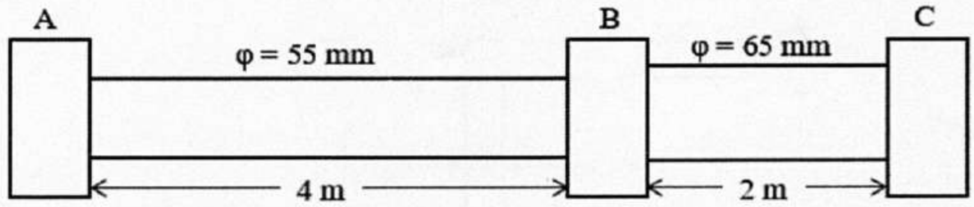
5(b). A rigid horizontal bar of negligible mass is connected to two rods as shown in figure. If the system is initially stress-free, calculate the temperature change that will causes a tensile stress of 90 MPa in the brass rod. Assume that both rods are subjected to the change in temperature. 15



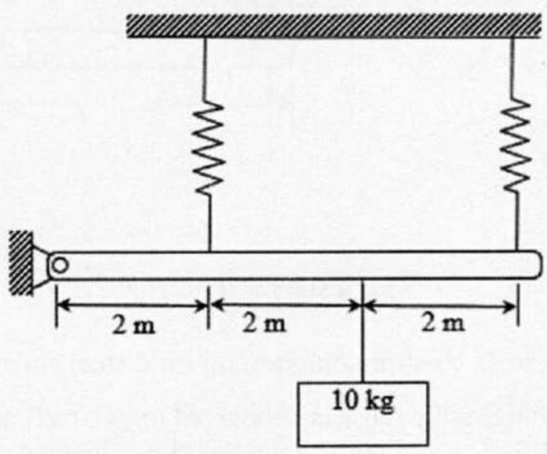
5(c). The composite rod shown in figure is stress free before the axial load P_1 and P_2 are applied. Assuming that the walls are rigid, calculate the stress in each material if P_1 = 150 kN and P_2 = 90 kN. 15



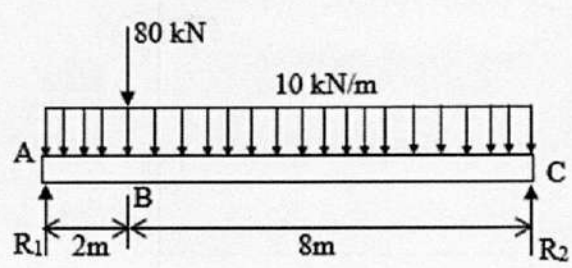
6(a). The steel shaft shown in figure rotates at 4 Hz with 35 kW taken off at A, 20 kW is removed at B and 55 kW is applied at C. Using E = 83 GPa, find the maximum shear stress and the angle of rotation of gear A relative to gear C. 17



6(b). A rigid bar hinged at one end, is supported by two identical springs as shown in figure. Each spring consists of 20 turns of 10 mm wire having a mean diameter of 150 mm. Compute the maximum shearing stress in the springs. Neglect the mass of the rigid bar. 18



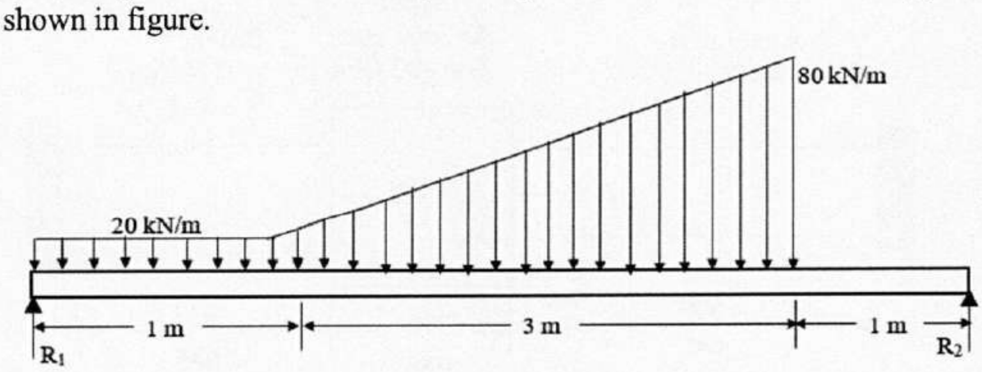
7(a). Write shear and moment equation for the beam loaded as shown in figure and sketch the shear and moment diagram. 18



7(b). Two C 310 × 45 channels are latticed together, so they have equal moment of inertia about the principle axes. Determine the minimum length of a column having this section assuming pinned ends, E = 200 GPa and a proportional limit of 240 MPa. What safe load will the column carry for the length of 12 m with a factor of safety of 2.5. For C 310 × 45, take $I_x = 67.3 \times 10^{-6} m^4$. 17

8(a). Select the lightest W shape that will act as column of 8 m long with hinged ends and support an axial load of 270 kN with factor of safety of 2.5. Assume that, proportional limit is 200 MPa and E = 200 GPa. 17

8(b). Without writing shear and moment equations, draw shear and moment diagram for the beam shown in figure. 18



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2^{nd} Year 1st Term Examination, 2021
ME 2115
(Fluid Mechanics)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). What is a fluid? How does it differ from a solid? 07

1(b). What is viscosity? How does the kinematic viscosity of liquids and gases vary with temperature? 10

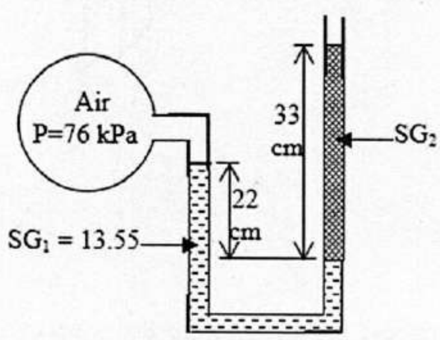
1(c). What is surface tension? What is its cause? 08

1(d). A capillary tube of 1.3 mm diameter is immersed vertically in water exposed to the atmosphere. Determine how high water will rise in the tube. Take the contact angle at the inner wall of the tube to be 6° and the surface tension to be 1.2 N/m. 10

2(a). What is the difference between gage pressure and absolute pressure? Prove that the pressure is the same in all directions at a point in a static fluid. 11

2(b). Explain how the resultant hydrostatic force and its direction can be determined on submerged curve surface? 11

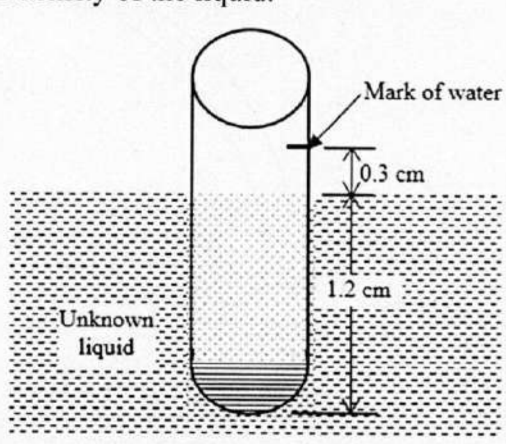
2(c). Consider a double-fluid manometer attached to an air pipe shown in figure. If the specific gravity of one fluid is 13.55, determine the specific gravity of the other fluid for the indicated absolute pressure of air. Take the atmospheric pressure to be 100 kPa. 13



3(a). How buoyant force is different from hydrostatic force? Show that buoyant force is independent of the distance of the body from the free surface. Explain why? 12

3(b). What is metacentre? How metacentric height is the measure of stability for floating body? Demonstrate the stabe and unstable configurations with neat sketch. 12

3(c). The density of a liquid is to be determined by an old 1 cm diameter cylindrical hydrometer whose division marks are completely wiped out. The hydrometer is first dropped in water, and water level is marked. The hydrometer is then dropped into the other liquid, and it is observed that the mark for water has risen 0.3 cm above the liquid-air interface as shown in figure. Determine the density of the liquid. 11



4(a). Explain the ways of visualization of fluid flow with appropriate diagrams. Derive the expression of streamline and explain how to draw streamline using this expression. 14

4(b). "A fluid element will rotate if there are net changes in velocity in flow direction" – justify the statement based on the concept of vorticity. Deduce the condition of irrotationality for 3-D flow. 11

4(c). A steady, incompressible, two-dimensional (in the xy-plane) velocity field is given by $\vec{V} = (0.523 - 1.88x + 3.94y)\hat{\imath} + (-2.44 + 1.26x + 1.88y)\hat{\jmath}$. Calculate the acceleration at the point $(x, y) = (-1.55, 2.07)$. 10

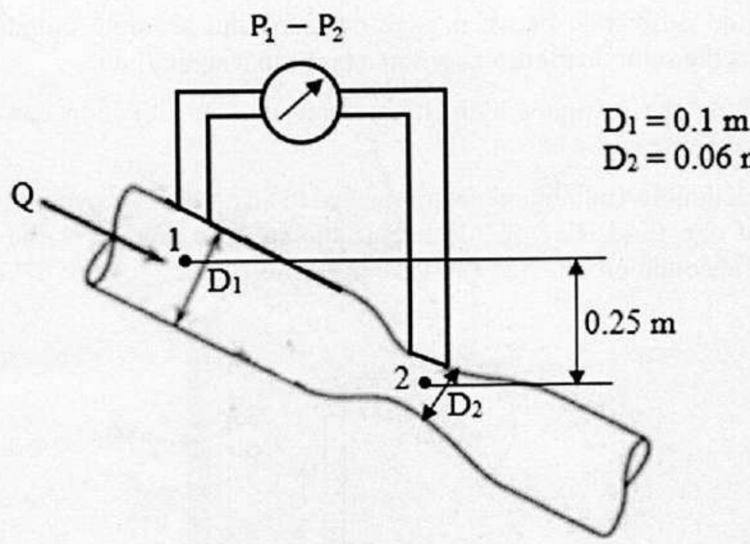
SECTION – B

5(a). Derive the differential form of the continuity equation for steady incompressible flow. 13

5(b). What is stagnation pressure? Explain how it can be measured. 05

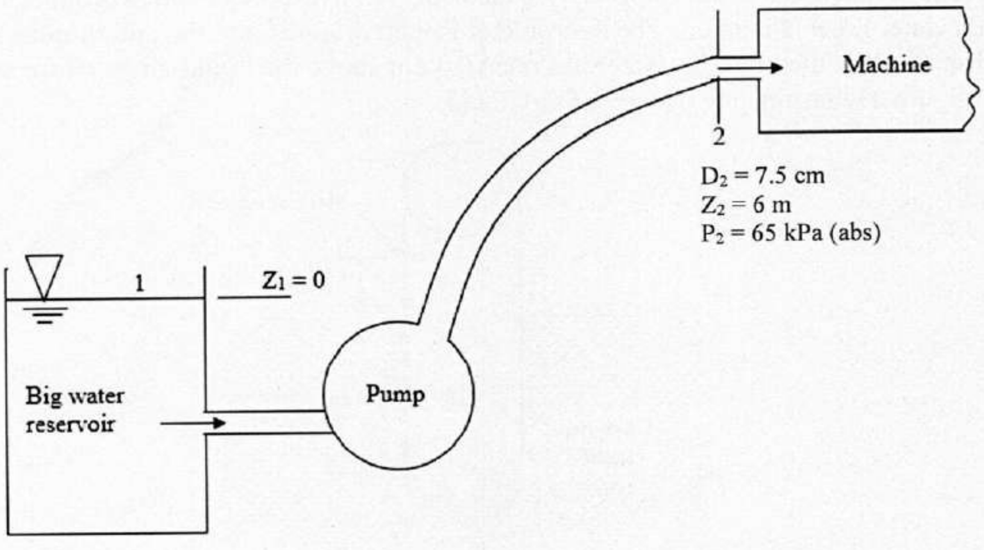
5(c). What is hydraulic grade line? How does it differ from the energy grade line? 05

5(d). Kerosene (SG = 0.85) flow through the venturimeter. The difference in pressure in the gage is 120 kPa as shown in figure. Determine the flow rate. 12



6(a). How the Reynolds transport theorem relate the system approach and control volume approach? Derive the expression for this theorem and use it to deduce the mass conservation equation in integral form. 20

6(b). A pump delivers water at 0.04 m^3/s to a machine at section 2 which is 6 m higher than the reservoir surface. The losses between section 1 and 2 are given by $h_f = \frac{KV_2^2}{2g}$, where K = 7.5 is a dimensionless loss coefficient. Find the horsepower required for the pump if it is 80% efficient. Take α = 1.07. 15



7(a).	What does boundary layer displacement and momentum thickness physically represent for external viscous flow? Explain with sketches and how do the thickness can be calculated?	15
7(b).	Derive the expression for the total friction force on plate of length L and width b. Consider 2-D laminar boundary layer flow and assume the velocity profile in the boundary layer is sinusoidal as follows: $\frac{u}{U} = \sin\left(\frac{\pi}{2} - \frac{y}{\gamma}\right)$ The symbols have their usual meanings.	20
8(a).	Why boundary layer developed for flow over an object?	04
8(b).	Explain the terms (i) drag force, and (ii) lift force.	08
8(c).	What is flow separation? What causes it? What is the effect of flow separation on the drag coefficient?	08
8(d).	A sharp flat plate with L = 50 cm and b = 3 m is parallel to a stream of velocity 2.5 m/s. Find the drag on one side of the plate, and the boundary thickness γ at the trailing edge for the followings: (i) air ($\rho = 1.2\ \text{kg/m}^3$, $\nu = 1.5 \times 10^{-5}\ \text{m}^2/\text{s}$) (ii) water at 20°C ($\rho = 998\ \text{kg/m}^3$, $\nu = 1.005 \times 10^{-6}\ \text{m}^2/\text{s}$	15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2021

CSE 2113

(Computer Programming)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). What is GNU Octave? What is GNU stand for? Write the application and limitation of GNU octave, Compare octave and MATLAB. 13

1(b). What are the different data types used in MATLAB language? What data types can store data from different classes? Give example of them. 09

1(c) What is the purpose of using assignment operator in octave? Write down the rules that you must to follow during the naming of variable. 13

2(a). What is mean by script file.Why do we need script file explain with an example. 10

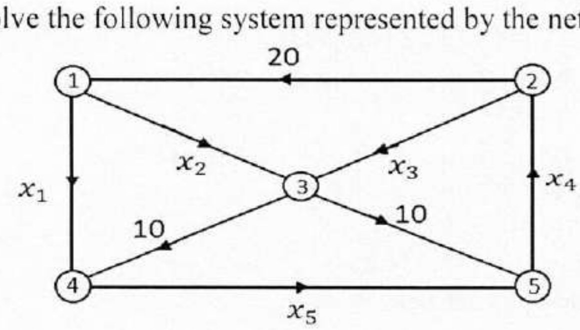
2(b). Write a octave script to calculate the value of k 12

$$k = \sigma\sqrt{\pi\alpha}\left[\frac{1-\frac{a}{2b}+0.326(\frac{a}{b})^2}{\sqrt{1-\frac{a}{b}}}\right]$$

for $\sigma = 1200$ psi, h=5 in , b= 4 in. and a=1.5 in.

2(c) What are the steps you need to follow to save the output of your program to a file using the `f print f` command? Explain. 13

3(a). Write octave code to solve the following system represented by the network shown in figure 12



i) solve the system by considering augmented matrix
ii) solve the system by considering coefficient matrix.

3(b). Explain matrix decomposition with example: For the following matrix, write MATLAB code to calculate i) LU decomposition of A, and ii) SVD of A 11

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -2 & 5 & -2 \\ -6 & 6 & -3 \end{bmatrix}$$

3(c). Define Eigen value and Eigen vectors with geometric interpretation of Eigen values. Using octave commands find the Eigen values and Eigen vectors of the following Matrix T. 12

$$T = \begin{bmatrix} -1 & 0 & -1 \\ -1 & -3 & 0 \\ -4 & -13 & -1 \end{bmatrix}$$

4(a)	What is meant by 'function'? Why do we need 'function'? Explain user defined of an Octave function with example?	14
4(b).	Write a user defined function in Octave for the function	14

$$y = f(x) = \frac{x^4\sqrt{3x+5}}{(x^2+1)^2}$$

The input to the function is x and the output is y Use the function to calculate y for x=1, 3,5,7,9 and 11.

i) using anonymous function named as `'myfunctiona'`

ii) using function file named as `'myfunctionf'`

4(c). Suppose a user defined function A contains two nested functions B and C at the same level. How will you define this function in Octave? Write down Pseudo-code. 07

SECTION – B

5(a). Explain the function of the commands with syntax and example: 15

i) repmat ii) find iii) fliplr iv) flipud v) logspace

5(b). Write down the output of following octave command. 07

i)
```
x=1:-9.50
a=length(x)
```

ii)
```
b=[2,3,4,5,6,7,8]
b(4)
```

iii)
```
b=[2,3,4,5,6,7,8]
c=b'
d=b(1,4)
```

5(c). An octave code is written as : 06

```
y=[-1,6,15,-7,31,2,-4,-5]
x=y(3:5)
x1 =(y(1:2),y(7:8))
x2 =(y(1),y(2),y(7),y(8))
Y=y1
n=y(4,1)
x3 = y'
n1 =y(1,4)
index=(1,2,7,8)
Z=y1(index)
[ynew,indx]=short(y)
[ynew,indx]=sort(y, descend)
```

Is there any error in the code? If yes, then find out the error and write down right code.

5(d). By writing a single line octave code create the following matrix. 07

i) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ii) $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ 1 & 1 & 0 & 3 \end{bmatrix}$ iii) $\begin{bmatrix} 32 & 32 & 32 \\ 32 & 0 & 32 \\ 32 & 32 & 32 \end{bmatrix}$

6(a). Why line specifiers is used in plotting? When will you use figure command? 10

6(b). Plot the function $f(x) = x^3 - 2x^2 - 10\sin^2\text{x} - e^{0.9x}$ and $g(x) = -3x^4 + 10x^2 - 3$ for $-2 \le x \le 4$ in one figure. Plot f(x) with a solid line, and g(x) with a dashed line. Add a legend and label to the axes. 10

6(c). How will you use line command for plotting multiple graphs in the sample plot? Explain briefly. 07

6(d). A function is given as $z = r\theta$ where $0 \le \theta \le 360°$ and $0 \le r \le 2$. Write an octave code for followings: 8

i) To Plot the surface in 3-D plane.
ii) To take projection in xy, xz and yz plane.

7(a). What is the difference between roots and poly command? 05

7(b). The following polynomial can be used to relate the zero pressure specific heat of dry air c_p in kJ/(kg K) to temperature in K. 08

$c_p = 0.99403 + 1.671 \times 10^{-4}T + 9.7215 \times 10^{-8}T^2 - 9.5838 \times 10^{-11}T^3$.

write Octave code to determine :

i)c_p corresponds to a range of T=0 to 1200K

ii) The temperature that corresponds to a specific heat of 1.1 kJ/(kg K).

7(c). Mention the cases when the following function will not suitable for curve fitting. 05

i) Exponential, ii) Logarithmic, iii) Reciprocal.

7(d). Suppose $x = [20, 29]$ and $y = [100, 121]$. Write an octave code to determine the value of y at $x = 36$ through interpolation. 04

7(e). An experiment data fits the following equation given below: 13

$$\mu = \frac{CT^{3/2}}{T + S}$$

Through curve fitting how will you determine C and S? Explain.

Write the octave code for determining C and S for the data given in following table.

T	-20	0	40	100	200	300	400
μ	1.63	1.71	1.87	2.17	2.53	2.98	3.32

8(a). What is the main difference between 'for loop' and 'while loop'? 04

8(b). What are the steps you must follow to execute 'while loop' properly? 06

8(c). Write down the output of following codes: 16

i)
```
n=5;
for k=0:0.5:n
disp('ESE2K19')
if K>3.5
  break
end
end
```

ii)
```
n=5;
  for k=0:0.5:n
  if k> 3.5
    break
  end
  disp('ESE2K19')
  end
```

ii)
```
m=1;
while 1
disp('CSE2113')
m=m+1;
end
```

iv)
```
m=1
while 1
disp ('CSE 2113')
m=m+1;
if m>=10
   break
end
end
```

Also give a brief explanation of your answer.

8(d). For the given series 09

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$+-----------+$\frac{x^k}{k!}$+------

Use for loop to calculate $e^{1.5}$ using 20 terms of the series above.

......End......